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ABSTRACT

WealthWise AI

The WealthWise AI is an intelligent application designed to help individuals manage their finances through automated expense tracking, smart categorization, and predictive budgeting. It supports both static input through receipt scanning and dynamic input through real-time voice-based transaction logging using tablets and smartphones.

The project uses advanced machine learning methods to analyze spending patterns and generate actionable financial insights. It processes transaction data, extracts key features, and trains models using modules such as Keras, TensorFlow, NumPy, Pandas, scikit-learn, and Matplotlib. Natural language processing is handled through FinBERT and Stanza, while forecasting is powered by Facebook Prophet and LSTM networks.

This system also provides predictive analysis by forecasting future expenses based on historical data, helping users in effective budget planning. It offers features such as personalized financial recommendations, bill reminders, group expense management, and interactive dashboards for better visualization.

This system offers an easy and efficient approach to personal finance management and can be applied in areas like budget planning, expense monitoring, and financial decision-making.

Hardware Requirements

- * Processor – Intel Core i5 / i7 or equivalent
- * RAM – 8 GB or higher
- * Storage – 256 GB SSD or higher
- * Internet Connection – Required for voice and web features

Software Requirements

- * **Operating System** – Windows 10/11 or Linux (Ubuntu 22.04+)
- * **Programming Language** – Python 3.9+
- * **Frameworks** – Django, Tailwind CSS
- * **Libraries** – Keras, NumPy, Pandas, scikit-learn, Matplotlib, Stanza, Prophet, Tesseract OCR
- * **Task Queue** – Celery with Redis
- * **Database** – PostgreSQL
- * **Development Environment** – VS Code / PyCharm
- * **Additional Tools** – Microphone (for voice logging), Camera/Scanner (for receipt upload)